NDIS high cost AT report 를 작성해야해. MS 환자이고 바이오네스 펀딩이 필요해. 다른사람이 작성해놓은 레포트가 있으면 찾아봐

네. 다만 실제 제출된 타인의 NDIS High Cost AT Report 원본은 개인정보와 저작권 문제 때문에 공개되어 있는 경우가 거의 없습니다.

대신, 제가 찾아보니 참고할 만한 자료는 있습니다.

NDIS의 공식 High Cost AT 보고서 작성 기준
실제 호주 Orthotist/AOPA에서 소개한 MS + Bioness L300 Go 사례
호주 공급업체들이 사용하는 Bioness 임상 근거 자료

그리고 중요한 점은, NDIS 심사관은 기기 자체(Bioness)를 보는 것이 아니라 "왜 이 사람에게 반드시 필요한지"를 봅니다.

실제 심사관이 기대하는 High Cost AT Report 구성

1. Executive Summary (1 page)
   Participant diagnosis
   Current functional limitation
   AT trial outcome
   Recommended AT
   Cost
   Expected outcomes
2. Participant Background
   MS diagnosis
   Disease progression
   Current mobility status
   Environmental factors
   Existing supports
3. Functional Assessment
   Body structure/function impairment
   Foot drop
   Weakness
   Spasticity
   Fatigue
   Balance deficit
   Reduced endurance
   Activity limitation
   Reduced walking distance
   Increased falls/trips
   Reduced community participation
   Participation restriction
   Shopping
   Social activity
   Employment
   Family participation
4. Current Equipment and Why It Is Insufficient

예시)

Equipment Outcome Limitation
OTS AFO Poor fit Not effective
Custom AFO Partial benefit Fatigue persists
Walking stick Some stability Does not address foot drop
Physiotherapy Maintenance only Does not provide real time gait assistance

이 부분이 매우 중요합니다.

NDIS는 항상 묻습니다.

"Why can't existing supports meet the participant's needs?"

5. Bioness Trial Results (가장 중요한 섹션)
   In-clinic trial

Observed improvements:

Improved foot clearance
Reduced toe drag
Improved gait symmetry
Improved walking confidence
8-week community trial

Observed improvements:

Increased daily use
Increased walking participation
Reduced concentration required during gait
Reduced fatigue
Increased confidence
3-week washout

Observed deterioration after device removal:

Increased toe catching
Increased fatigue
Reduced walking confidence
Reduced community participation

이 washout 데이터는 상당히 강력한 근거가 됩니다.

6. Alternative Options Considered

이 섹션은 심사관이 굉장히 중요하게 봅니다.

Alternative Reason not recommended
Continue without AT Unsafe
Walking stick only Does not correct foot drop
Carbon fibre AFO Does not adapt to changing fatigue
Continue physiotherapy only Not sufficient
Bioness L300 Go Most effective
7. NDIS Reasonable and Necessary Criteria
Section 34 justification
A. Related to disability ✓

MS directly causes:

Foot drop
Weakness
Fatigue
Mobility impairment
B. Effective and beneficial ✓

Objective improvements demonstrated through:

Clinical trial
Community trial
Washout period

C. Value for money ✓

Although initial cost is high:

Potential reduction in:

Falls
Carer burden
Future support requirements
Injury risk

Long-term participation outcomes outweigh cost.

D. Most appropriate funding source ✓

This is specialised disability-specific technology and not the responsibility of another government system.

제가 추천하는 제출용 목차 (12~18페이지)
Cover Page
Executive Summary
Participant Information
Medical Background
Current Functional Status
Existing Supports
Clinical Assessment Findings
Goal Alignment
Alternative Options Considered
In-Clinic Trial Results
8-week Community Trial Results
3-week Washout Results
Risk Assessment
Cost Benefit Analysis
NDIS Reasonable and Necessary Criteria
Final Recommendation
Capacity Building Hours
Supporting Documents & Outcome Measures

그리고 사실 저는 지금 작성하려는 대상이 이전에 같이 작업했던 동일한 MS + Bioness L300 Go 환자라고 생각합니다.

(35세 MS, 우측 foot drop, 8주 커뮤니티 트라이얼 + 3주 washout 자료를 모아서 제출하려는 케이스)

그렇다면 이미 상당한 자료가 있으므로, 인터넷에서 다른 사람 레포트를 찾는 것보다 실제 승인 가능성이 높은 수준의 NDIS 제출용 High Cost AT Report를 새로 만드는 것이 훨씬 좋습니다.

원하시면 제가 **실제 Orthotist가 제출하는 15~20페이지 수준의 NDIS High Cost AT Report (Bioness L300 Go, MS 환자용)**를 처음부터 작성해드릴 수 있습니다.

뉴롤로지스트가 써준 레터야. 바이이오네스 금액은 40000불이 넘을거 같아 정확한 견적서는 아직 안만들었어
환자의 NDIS goal이야. 이걸 사용해서 위의 너가 수정해준 내용들 포함해서 실제 제출 가능한 20-25페이지짜리 최종 제출본 만들어줘
Mon, Jun 29 at 2:12 PM
NDIS_Full_AT_Report_Bioness_L300Go_M(1).docx
Document

맞습니다.

제가 방금 다시 작성한 버전은 Executive Summary 중심의 "짧은 최종본" 에 가깝습니다.

실제 NDIS High Cost AT Report에서 승인율이 높은 보고서는 보통 Executive Summary만 해도 3~4페이지 정도 되고, NDIS Delegate가 Block 1만 읽어도:

왜 필요한지
어떤 평가를 했는지
어떤 결과가 나왔는지
왜 다른 옵션은 부족한지
왜 NDIS 기준을 충족하는지

를 거의 이해할 수 있게 작성합니다.

그래서 사실 지금 보고서에서는 기존 Block 1 구조를 유지하는 것이 더 좋습니다.

제가 권장하는 방법

기존 Block 1 (길고 상세한 버전)

이번에 수정한

0/5 vs 3/5
기능 중심 논리
Neurologist 설명

만 반영

입니다.

즉,

❌ 새로 작성한 짧은 버전으로 교체

가 아니라

✅ 기존 Block 1에 일부 문장만 수정

입니다.

실제로 수정해야 할 부분
Executive Summary - Participant Overview

기존 문장

Clinical assessment identified complete loss of active ankle dorsiflexion (0/5), resulting in severe foot drop.

찾아서 삭제

아래 문장으로 교체

Clinical assessment identified severe dorsiflexor weakness resulting in persistent foot drop, reduced foot clearance and gait inefficiency during community mobility. Physiotherapy assessment documented dorsiflexion strength of 0/5, while subsequent neurological assessment identified active dorsiflexion of approximately 3/5. Despite this variation, both assessments confirmed clinically significant functional impairment affecting walking performance, endurance and participation.

Executive Summary - Assessment Findings

기존 문장

Complete loss of dorsiflexion contributed to significant foot drop.

찾아서 교체

수정

Significant dorsiflexor weakness contributed to persistent foot drop, reduced foot clearance and increased reliance on compensatory gait strategies.

Executive Summary - Neurologist Endorsement

기존 문장

The neurologist noted improvement from 3/5 to 4-/5 dorsiflexion strength.

그대로 두고

바로 아래 추가

Although measurable muscle activation was present during neurological assessment, functional gait impairments remained evident without the Bioness system. The recommendation is therefore based on demonstrated functional outcomes rather than muscle strength findings alone.

Clinical Conclusion

기존 결론 마지막에 추가

While active dorsiflexion was identified during neurological assessment, M continued to demonstrate clinically significant disability-related mobility limitations. The recommendation is therefore based on functional impairment, participation restrictions and objectively demonstrated trial outcomes rather than isolated muscle strength findings.

사실 더 중요한 수정

NDIS Delegate가 보는 관점에서는

0/5인지

3/5인지

생각보다 중요하지 않습니다.

오히려 Delegate는

"3/5면 왜 Bioness가 필요하지?"

를 봅니다.

그래서 보고서 전체에서 가장 중요한 문장은 여전히:

M does not require the Bioness L300 Go because she is unable to walk. Rather, she requires the device because she currently maintains mobility only through significant activity restriction, compensatory strategies and avoidance of meaningful activities.

입니다.

이 문장이 Executive Summary, Clinical Summary, Final Recommendation에 각각 한 번씩 들어가 있으면 훨씬 강력합니다.

제 생각에는 현재 작성된 Block 1~6 중에서 가장 승인 가능성을 높일 수 있는 부분은 Block 1 전체를 다시 쓰는 것이 아니라, Executive Summary를 1.5~2페이지 정도 더 확장하고 "왜 AFO가 아니라 Bioness인지"를 Executive Summary에 추가하는 것입니다. 그게 실제 NDIS High Cost AT 보고서에서 Delegate 설득력에 가장 큰 영향을 줍니다.

기존 블럭 1에서 너가 권장하는 방법의 수정사항을 수정해서 보여줘

좋습니다. 아래는 기존 상세 버전 Block 1을 유지하면서, 제가 권장한 수정사항(0/5 vs 3/5, 기능 중심 논리, Neurologist 설명 강화)을 반영한 교체 문구들입니다.

1. Executive Summary – Participant Overview
   기존 문장 삭제

Clinical assessment identified complete loss of active ankle dorsiflexion (0/5), resulting in severe foot drop during gait.

아래 문장으로 교체

Clinical assessment identified significant dorsiflexor weakness resulting in persistent foot drop, reduced foot clearance and gait inefficiency during community mobility. Physiotherapy assessment documented dorsiflexion strength of 0/5, while subsequent neurological assessment identified active dorsiflexion of approximately 3/5. Despite this variation in manual muscle testing, both assessments confirmed clinically significant functional impairment affecting walking performance, endurance, safety and participation.

2. Executive Summary – Assessment Findings
   기존 문장 삭제

Complete loss of dorsiflexion contributed to significant foot drop and reduced mobility.

아래 문장으로 교체

Significant dorsiflexor weakness contributed to persistent foot drop, reduced foot clearance, increased energy expenditure and reliance on compensatory gait strategies. These impairments became more pronounced with fatigue and significantly impacted community mobility and participation in meaningful activities.

3. Executive Summary – Participant Overview 마지막 문단에 추가

현재 문단 끝에 아래 문장 추가

Although active dorsiflexion was identified during neurological assessment, functional gait impairments remained evident. M continued to demonstrate reduced foot clearance, fatigue-related gait deterioration and activity restriction, indicating that muscle strength alone was insufficient to overcome her disability-related mobility limitations.

4. Neurologist Endorsement Section 수정

현재 문장

The neurologist also reported improved active ankle dorsiflexion during the trial period.

바로 아래 추가

Although measurable muscle activation was present during neurological assessment, clinically significant gait impairments persisted without the Bioness system. The participant continued to experience reduced walking efficiency, reduced endurance and increased activity restriction when the device was not available. This indicates that the recommendation is based on demonstrated functional outcomes rather than muscle strength findings alone.

5. Washout Findings Section 마지막 문단 추가

현재 마지막 문장

The deterioration observed during washout confirms that the improvements achieved during the trial were directly attributable to the Bioness system.

바로 아래 추가

Importantly, the deterioration observed during washout occurred despite the presence of active dorsiflexion documented during neurological assessment. This further supports the conclusion that isolated muscle strength findings do not adequately reflect the participant's functional mobility limitations and participation restrictions.

6. Clinical Conclusion Section 수정

현재 문장

M does not require the Bioness L300 Go because she is unable to walk.

Rather, she requires the device because she currently maintains mobility only through significant activity restriction, compensatory strategies and avoidance of meaningful activities.

바로 아래 추가

While active dorsiflexion was identified during neurological assessment, M continued to demonstrate clinically significant disability-related mobility limitations. The recommendation is therefore based on functional impairment, participation restrictions, fatigue-related gait deterioration and objectively demonstrated trial outcomes rather than isolated muscle strength findings.

7. Recommendation Section 마지막 강화 문장 추가

현재 마지막 문단 뒤에 추가

The recommendation is not based on a diagnosis of Multiple Sclerosis or a specific muscle strength grade. Rather, it is based on the participant's demonstrated inability to sustain meaningful community mobility and participation without the assistance of the Bioness system. The community trial and subsequent washout period clearly demonstrated that ongoing access to the device is necessary to maintain the functional gains achieved during the trial.

가장 중요한 추가 문장 (Block 1에서 반드시 2회 사용)

Executive Summary 중간과 Clinical Conclusion에 아래 문장을 넣는 것을 권장합니다.

M does not require the Bioness L300 Go because she is unable to walk. Rather, she requires the device because she currently maintains mobility only through significant activity restriction, compensatory strategies and avoidance of meaningful activities.

이 문장이 0/5 vs 3/5 논란을 사실상 무력화시키고, NDIS 심사관의 초점을 근력 수치가 아닌 기능적 필요성 (functional need) 으로 옮겨주는 핵심 문장입니다.

위의 내용을 반영해서 블럭 1 작성해줘

BLOCK 1 – EXECUTIVE SUMMARY (FINAL SUBMISSION VERSION)
COVER PAGE
High Cost Assistive Technology Assessment Report
Bioness L300 Go Functional Electrical Stimulation System

Prepared for submission to the National Disability Insurance Scheme (NDIS)

Participant: M

NDIS Number: [Insert]

Date of Birth: [Insert]

Primary Diagnosis: Multiple Sclerosis

Assistive Technology Requested: Bioness L300 Go Functional Electrical Stimulation System (Lower Leg Component and Thigh Component)

Prepared By: [Orthotist Name], CO

Organisation: KOrthotics

Date of Report: [Insert]

TABLE OF CONTENTS

1. EXECUTIVE SUMMARY
   Purpose of Report

This report has been prepared to support funding approval for a Bioness L300 Go Functional Electrical Stimulation (FES) System, including both the Lower Leg Component and Thigh Component, under the National Disability Insurance Scheme (NDIS).

The recommendation is based upon:

Comprehensive orthotic assessment
Functional mobility assessment
In-clinic Functional Electrical Stimulation trial
Eight-week community trial
Three-week washout period
Standardised outcome measures
Participant feedback
Carer feedback
Neurologist endorsement
Current evidence-based practice

The purpose of the recommended assistive technology is to address disability-related mobility limitations resulting from Multiple Sclerosis and to support improved participation, independence, safety and quality of life.

Participant Overview

M is an adult woman living with Multiple Sclerosis who experiences significant functional limitations associated with neurological weakness, spasticity, fatigue and gait dysfunction.

Although M remains independently ambulant with the assistance of a four-wheeled walker, her mobility is maintained through substantial compensatory effort, activity restriction and avoidance of meaningful activities.

Clinical assessment identified:

Significant dorsiflexor weakness
Persistent foot drop
Dynamic inversion during swing phase
Knee hyperextension during stance
Reduced walking endurance
Fatigue-related deterioration in gait quality
Reduced standing tolerance
Reduced confidence during community mobility

Physiotherapy assessment documented right ankle dorsiflexion strength of 0/5, while subsequent neurological assessment identified active dorsiflexion of approximately 3/5. Despite this variation in manual muscle testing, both assessments confirmed clinically significant functional impairment affecting walking performance, endurance, safety and participation.

Although active dorsiflexion was identified during neurological assessment, functional gait impairments remained evident. M continued to demonstrate reduced foot clearance, fatigue-related gait deterioration and activity restriction, indicating that muscle strength alone was insufficient to overcome her disability-related mobility limitations.

As a result of these impairments, M experiences substantial restrictions in community access, prolonged standing activities, participation in family outings and engagement in meaningful everyday activities.

Impact of Disability

The impact of M's disability extends well beyond walking mechanics.

To manage instability, fatigue and fear of falling, M has progressively modified her behaviour and reduced participation in many aspects of daily life.

Activities commonly affected include:

Community access
Shopping
Family outings
Attendance at appointments
Meal preparation
Cooking
Baking
Walking on uneven surfaces
Prolonged standing activities

Importantly, M has developed significant activity avoidance behaviours as a strategy to manage mobility challenges.

Consequently, the absence of frequent falls should not be interpreted as evidence of functional safety. Rather, it reflects the extent to which M restricts activity and participation in order to minimise risk.

M does not require the Bioness L300 Go because she is unable to walk.

Rather, she requires the device because she currently maintains mobility only through significant activity restriction, compensatory strategies and avoidance of meaningful activities.

Clinical Assessment Findings

Comprehensive assessment identified a combination of neurological, biomechanical and functional impairments contributing to M's reduced mobility and participation.

Key findings included:

Neurological Impairments
Significant dorsiflexor weakness
Foot drop
Dynamic inversion during swing
Moderate plantarflexor spasticity
Mild invertor spasticity
Biomechanical Impairments
Reduced foot clearance
Toe drag
Knee hyperextension
Reduced gait efficiency
Increased energy expenditure during walking
Functional Impairments
Reduced endurance
Reduced standing tolerance
Reduced confidence
Activity avoidance
Participation restrictions

Assessment findings demonstrated that M's difficulties extended beyond isolated foot drop and significantly impacted her ability to safely and effectively participate in daily life.

Alternative Supports Considered

A range of alternative interventions were considered as part of the assessment process, including:

Physiotherapy
Four-wheeled walker
Off-the-shelf Ankle Foot Orthosis
Custom Ankle Foot Orthosis

While these interventions provided varying levels of benefit, none adequately addressed the combined effects of foot drop, knee hyperextension, fatigue-related gait deterioration and participation restrictions.

An off-the-shelf AFO had previously been trialled and was unsuccessful.

A custom AFO was considered; however, it would provide passive positioning only and would not address the neurological impairments contributing to gait dysfunction in the same manner as Functional Electrical Stimulation.

Consequently, a Functional Electrical Stimulation trial was considered clinically appropriate.

In-Clinic Trial Findings

M completed an in-clinic trial of the Bioness L300 Go Lower Leg Component.

Immediate improvements were observed in:

Foot clearance
Swing phase mechanics
Walking efficiency
Stability
Walking confidence

Both the participant and treating clinicians reported meaningful improvements in gait quality compared with walking without the device.

These findings supported progression to a structured community-based trial.

Community Trial Outcomes

Following the successful clinic trial, M completed an eight-week community trial using both the Lower Leg Component and Thigh Component.

The trial evaluated device performance across a broad range of real-world environments including:

Home environments
Community settings
Medical appointments
Shopping centres
Family outings
Uneven terrain
Recreational activities

Objective outcome measures demonstrated clinically meaningful improvements across multiple domains.

Outcome Measure	Baseline	End Trial	Change
Timed Up and Go	42.57 sec	33.11 sec	22% Improvement
10 Metre Walk Test	35.19 sec	29.31 sec	17% Improvement
6 Minute Walk Test	57m	90m	58% Improvement
Dynamic Gait Index	11/24	14/24	Improved
30 Second Sit-to-Stand	9	10	Improved

The most significant improvement was observed in walking endurance, with a 58% increase in walking distance achieved during the Six-Minute Walk Test.

Collectively, these findings demonstrate improved mobility, improved efficiency of movement and increased functional capacity during device use.

Functional and Participation Outcomes

The most meaningful outcomes observed during the community trial extended beyond standardised outcome measures.

During active use of the Bioness system, M reported:

Increased confidence when walking
Improved standing tolerance
Increased participation in family activities
Increased community access
Reduced mental effort associated with walking
Reduced activity avoidance

Most significantly, M reported:

“I got to cook and bake like I used to.”

She also stated:

“I stopped overthinking every step I took. I felt stable. I felt happy.”

These statements demonstrate improvements not only in mobility but also in participation, wellbeing and quality of life.

Neurologist Endorsement

M's treating neurologist independently reviewed her progress during the trial period.

The neurologist documented:

Improved mobility
Improved endurance
Improved confidence
Improved quality of life

The neurologist also reported improvement in active ankle dorsiflexion during the trial period.

Although measurable muscle activation was present during neurological assessment, clinically significant gait impairments persisted without the Bioness system. The participant continued to experience reduced walking efficiency, reduced endurance and increased activity restriction when the device was not available.

This indicates that the recommendation is based on demonstrated functional outcomes rather than muscle strength findings alone.

Washout Findings

Following completion of the community trial, the Bioness system was returned and M completed a structured three-week washout period.

The washout findings provided some of the strongest evidence supporting the recommendation.

Following removal of the device, M experienced:

Reduced walking confidence
Increased fatigue
Reduced participation
Reduced walking activity
Increased activity avoidance
Return of foot drag
Return of gait instability

Four trip or near-fall incidents were documented during the washout period.

In contrast, no falls were reported during the active trial despite substantially increased activity levels and community participation.

This should be interpreted in the context of M's significant activity restriction and avoidance behaviours prior to the trial.

Importantly, the deterioration observed during washout occurred despite the presence of active dorsiflexion documented during neurological assessment. This further supports the conclusion that isolated muscle strength findings do not adequately reflect the participant's functional mobility limitations and participation restrictions.

The deterioration observed during washout confirms that the improvements achieved during the trial were directly attributable to the Bioness system.

Clinical Conclusion

M does not require the Bioness L300 Go because she is unable to walk.

Rather, she requires the device because she currently maintains mobility only through significant activity restriction, compensatory strategies and avoidance of meaningful activities.

The community trial demonstrated measurable improvements in:

Functional mobility
Walking endurance
Standing tolerance
Confidence
Community participation
Safety
Quality of life

While active dorsiflexion was identified during neurological assessment, M continued to demonstrate clinically significant disability-related mobility limitations.

The recommendation is therefore based on functional impairment, participation restrictions, fatigue-related gait deterioration and objectively demonstrated trial outcomes rather than isolated muscle strength findings.

The subsequent washout period confirmed that these benefits were not maintained following removal of the device.

Recommendation Summary

Based on comprehensive assessment findings, objective outcome measures, participant outcomes, specialist endorsement and washout evidence, it is my professional opinion that funding should be approved for:

Bioness L300 Go Functional Electrical Stimulation System

Including:

Lower Leg Component
Thigh Component
Initial Assessment and Fitting
Clinical Programming
Clinical Reviews
Device Optimisation
Ongoing Clinical Support

The recommendation is not based on a diagnosis of Multiple Sclerosis or a specific muscle strength grade.

Rather, it is based on the participant's demonstrated inability to sustain meaningful community mobility and participation without the assistance of the Bioness system.

The community trial and subsequent washout period clearly demonstrated that ongoing access to the device is necessary to maintain the functional gains achieved during the trial.

Accordingly, the requested assistive technology is considered reasonable and necessary under Section 34 of the NDIS Act 2013 and represents the most appropriate intervention currently available to address M's disability-related functional limitations and participation restrictions.